

Silicon P Channel MOS Type (U-MOSII)/Silicon Epitaxial Schottky Barrier Diode

SSM5G04TU

DC-DC Converter

- Combined Pch MOSFET and Schottky Diode into one Package.
- Low RDS (ON) and Low VF

Absolute Maximum Ratings (Ta = 25°C) MOSFET

Characteristics	Symbol	Rating	Unit
Drain-Source voltage	V _{DS}	-12	V
Gate-Source voltage	V _{GSS}	±12	V
Drain current	DC	I _D	A
	Pulse	I _{DP} (Note 2)	
Drain power dissipation	P _D (Note 1)	0.5	W
	t = 10s	0.8	
Channel temperature	T _{ch}	150	°C

Absolute Maximum Ratings (Ta = 25°C) SCHOTTKY DIODE

Characteristics	Symbol	Rating	Unit
Maximum (peak) reverse voltage	V _{RM}	15	V
Reverse voltage	V _R	12	V
Average forward current	I _O	0.5	A
Peak one cycle surge forward current (non-repetitive)	I _{FSM}	2 (50 Hz)	A
Junction temperature	T _j	125	°C

Absolute Maximum Ratings (Ta = 25°C) MOSFET, DIODE COMMON

Characteristics	Symbol	Rating	Unit
Storage temperature	T _{stg}	-55~125	°C
Operating temperature	T _{opr} (Note 3)	-40~85	°C

Note: Using continuously under heavy loads (e.g. the application of high temperature/current/voltage and the significant change in temperature, etc.) may cause this product to decrease in the reliability significantly even if the operating conditions (i.e. operating temperature/current/voltage, etc.) are within the absolute maximum ratings.

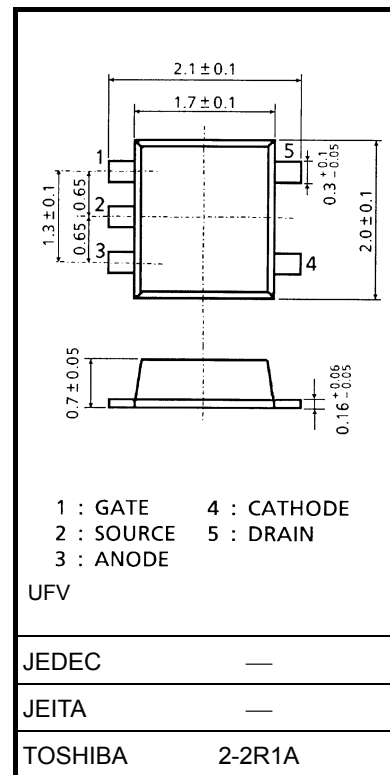
Please design the appropriate reliability upon reviewing the Toshiba Semiconductor Reliability Handbook ("Handling Precautions"/"Derating Concept and Methods") and individual reliability data (i.e. reliability test report and estimated failure rate, etc).

Note 1: Mounted on FR4 board
(25.4 mm × 25.4 mm × 1.6 t, Cu pad: 645 mm²)

Note 2: The pulse width limited by max channel temperature.

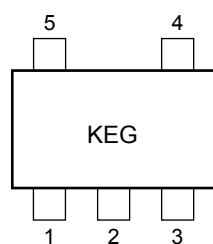
Note 3: Operating temperature limited by max channel temperature and max junction temperature.

Unit: mm

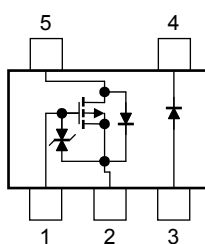


Weight: 7 mg (typ.)

Marking



Equivalent Circuit



Handling Precaution

When handling individual devices (which are not yet mounting on a circuit board), be sure that the environment is protected against electrostatic discharge. Operators should wear anti-static clothing and use containers and other objects that are made of anti-static materials.

The Channel-to-Ambient thermal resistance $R_{th(ch-a)}$ and the drain power dissipation P_D vary according to the board material, board area, board thickness and pad area. When using this device, please take heat dissipation fully into account.

MOSFET

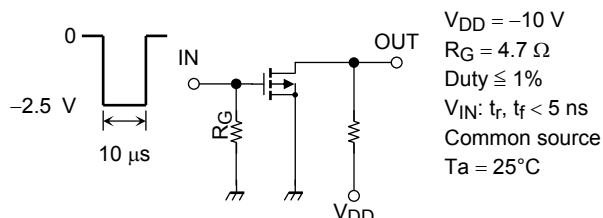
Electrical Characteristics (Ta = 25°C)

Characteristic	Symbol	Test Condition	Min	Typ.	Max	Unit
Gate leakage current	I_{GSS}	$V_{GS} = \pm 10 \text{ V}, V_{DS} = 0$	—	—	± 1	μA
Drain-Source breakdown voltage	$V_{(BR) DSS}$	$I_D = -1 \text{ mA}, V_{GS} = 0$	-12	—	—	V
	$V_{(BR) DSX}$	$I_D = -1 \text{ mA}, V_{GS} = +8 \text{ V}$	-4	—	—	
Drain Cut-off current	I_{DSS}	$V_{DS} = -12 \text{ V}, V_{GS} = 0$	—	—	-1	μA
Gate threshold voltage	V_{th}	$V_{DS} = -3 \text{ V}, I_D = -0.1 \text{ mA}$	-0.4	—	-1.1	V
Forward transfer admittance	$ Y_{fs} $	$V_{DS} = -3 \text{ V}, I_D = -0.5 \text{ A}$ (Note 4)	0.85	1.75	—	S
Drain-Source ON resistance	$R_{DS(ON)}$	$I_D = -0.5 \text{ A}, V_{GS} = -4 \text{ V}$ (Note 4)	—	200	240	$\text{m}\Omega$
		$I_D = -0.5 \text{ A}, V_{GS} = -2.5 \text{ V}$ (Note 4)	—	320	420	
Input capacitance	C_{iss}	$V_{DS} = -10 \text{ V}, V_{GS} = 0, f = 1 \text{ MHz}$	—	170	—	pF
Reverse transfer capacitance	C_{rss}	$V_{DS} = -10 \text{ V}, V_{GS} = 0, f = 1 \text{ MHz}$	—	32	—	pF
Output capacitance	C_{oss}	$V_{DS} = -10 \text{ V}, V_{GS} = 0, f = 1 \text{ MHz}$	—	58	—	pF
Switching time	Turn-on time	t_{on}	$V_{DD} = -10 \text{ V}, I_D = -0.5 \text{ A}$		18	ns
	Turn-off time	t_{off}	$V_{GS} = 0 \sim -2.5 \text{ V}, R_G = 4.7 \Omega$		14	

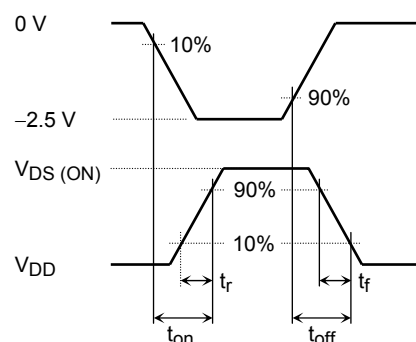
Note 4: Pulse measurement

Switching Time Test Circuit

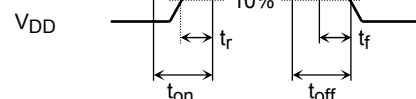
(a) Test circuit



(b) V_{IN}



(c) V_{OUT}



Precaution

V_{th} can be expressed as voltage between gate and source when low operating current value is $I_D = -100 \mu\text{A}$ for this product. For normal switching operation, $V_{GS(ON)}$ requires higher voltage than V_{th} and $V_{GS(OFF)}$ requires lower voltage than V_{th} .

(Relationship can be established as follows: $V_{GS(OFF)} < V_{th} < V_{GS(ON)}$)

Please take this into consideration for using the device.

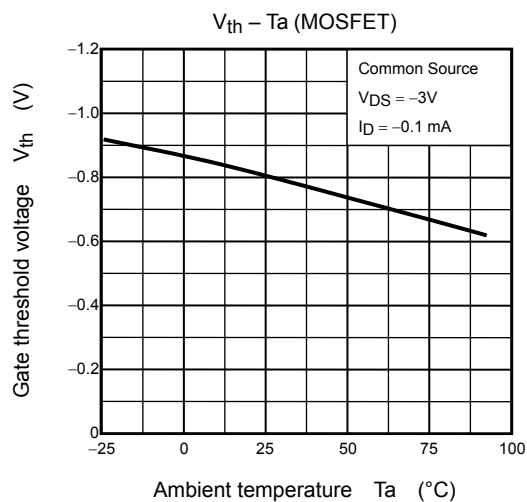
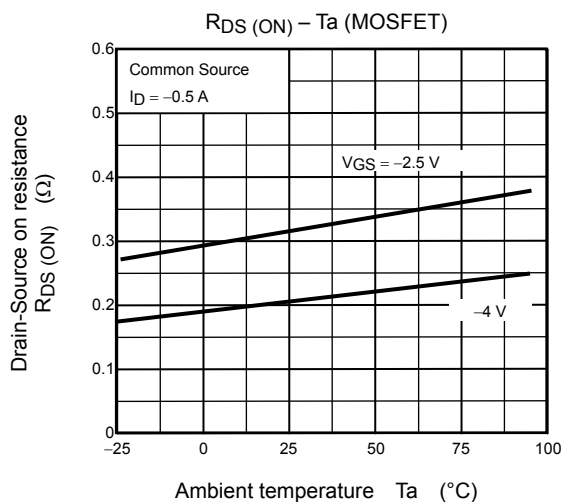
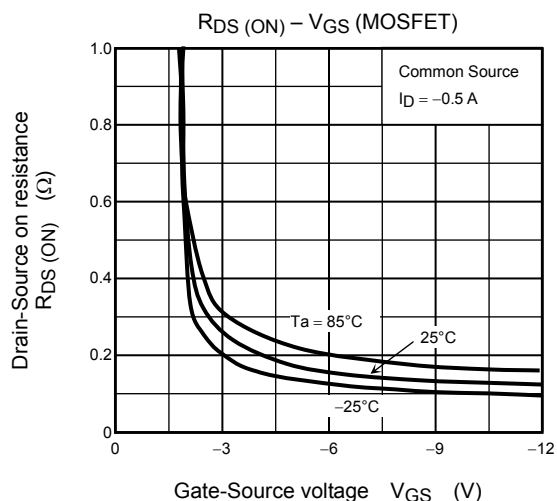
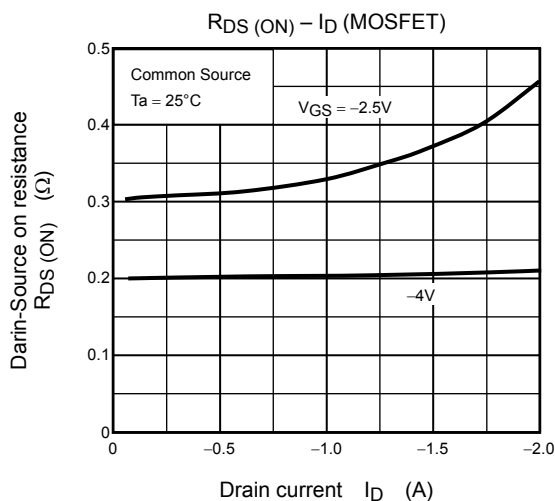
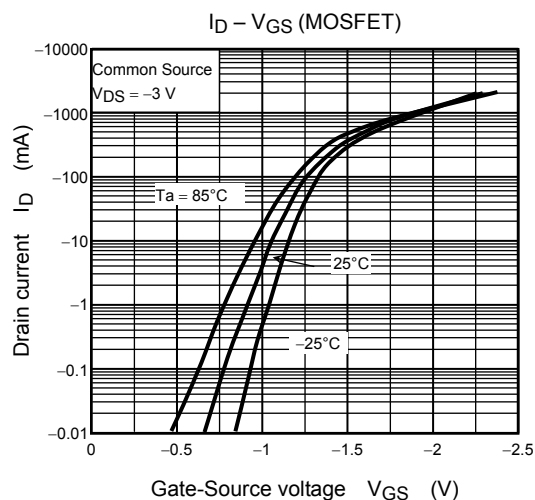
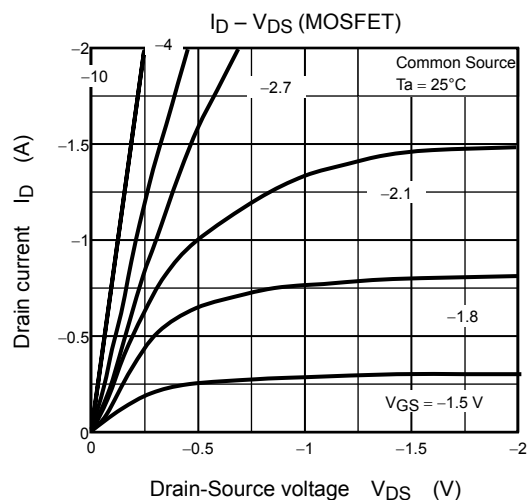
Schottky Diode**Electrical Characteristics (Ta = 25°C)**

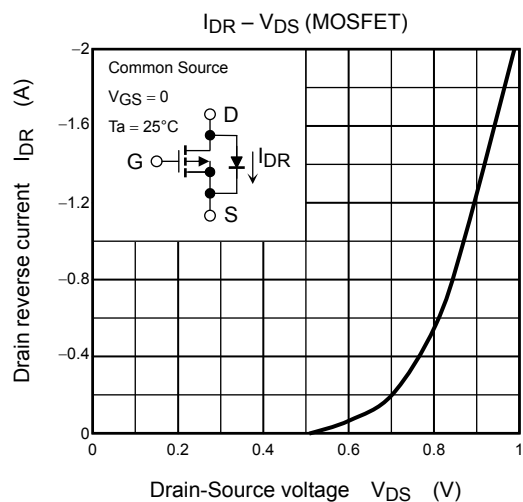
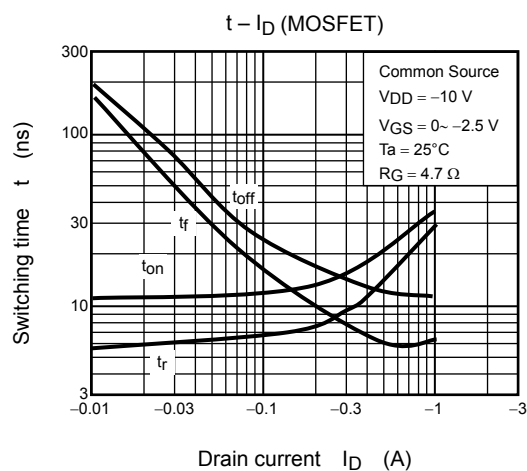
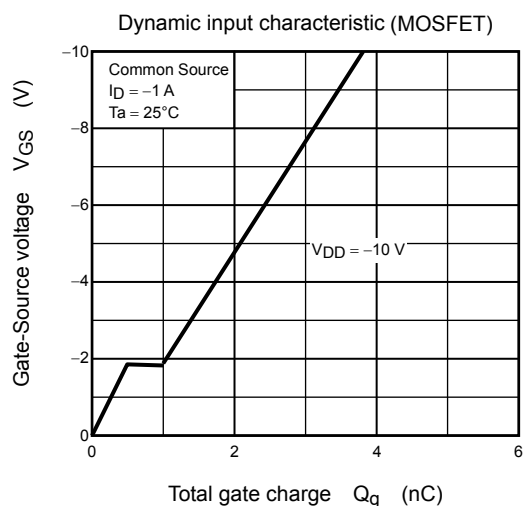
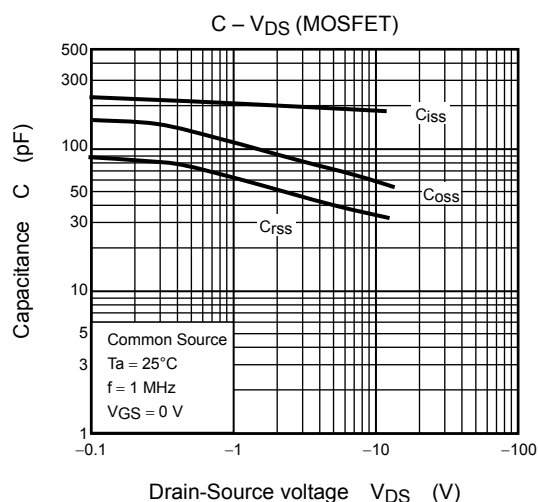
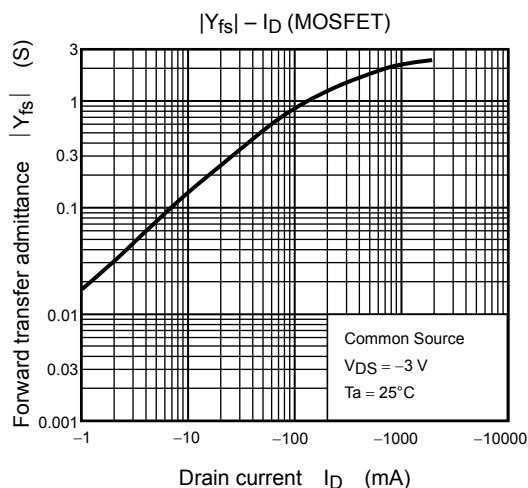
Characteristic	Symbol	Test Condition	Min	Typ.	Max	Unit
Forward voltage	V _{F (1)}	I _F = 0.3 A	—	0.33	0.39	V
	V _{F (2)}	I _F = 0.5 A	—	0.37	0.43	V
Reverse current	I _R	V _R = 12 V	—	—	100	μA
Total capacitance	C _T	V _R = 0 V, f = 1 MHz	—	80	—	pF

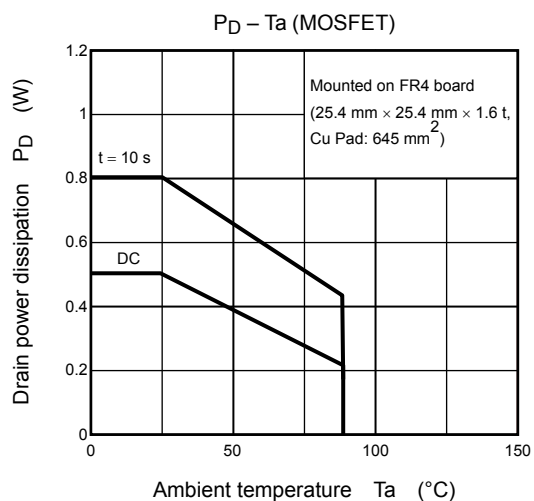
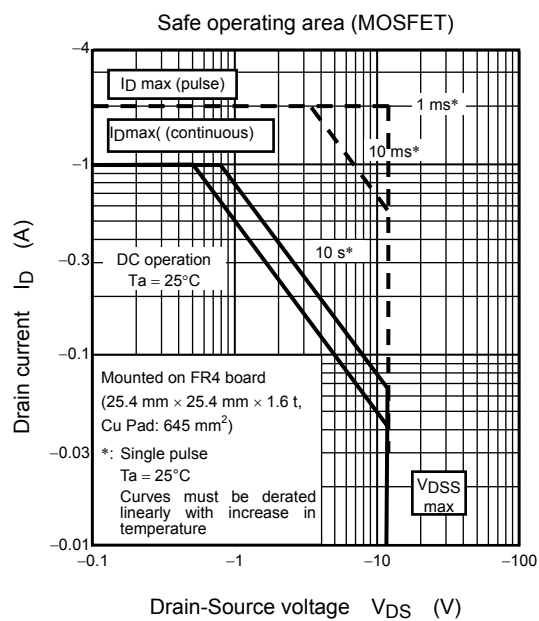
Precaution

The schottky barrier diode of this product are having large-reverse-current-leakage characteristic compare to the other switching diodes. This current leakage and not proper operating temperature or voltage may cause thermal runaway. Please take forward and reverse loss into consideration when you design.

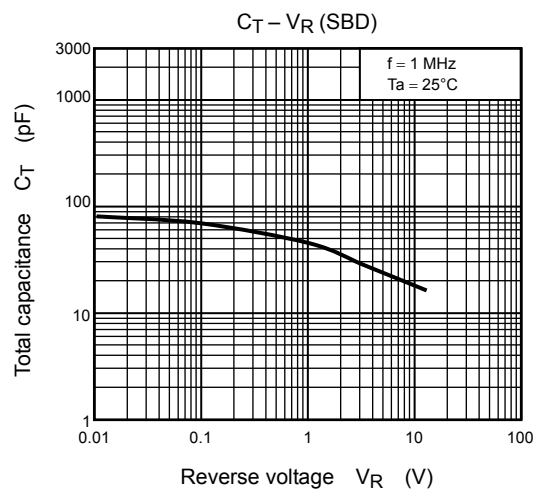
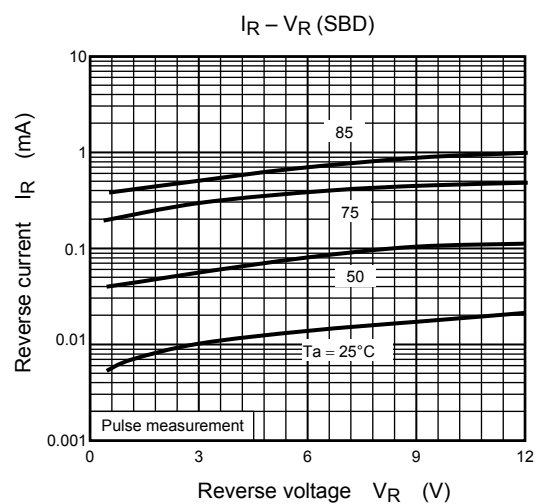
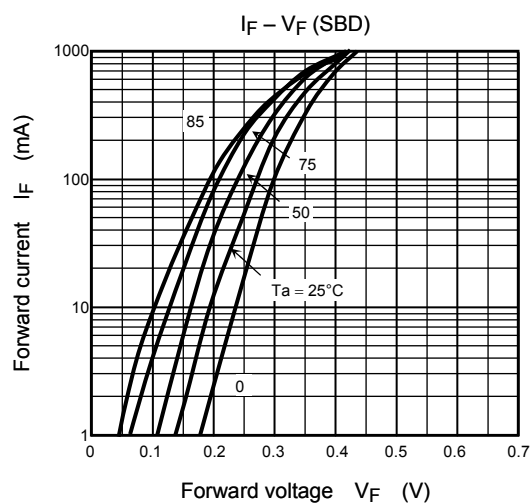
MOSFET Electrical Characteristics Graph



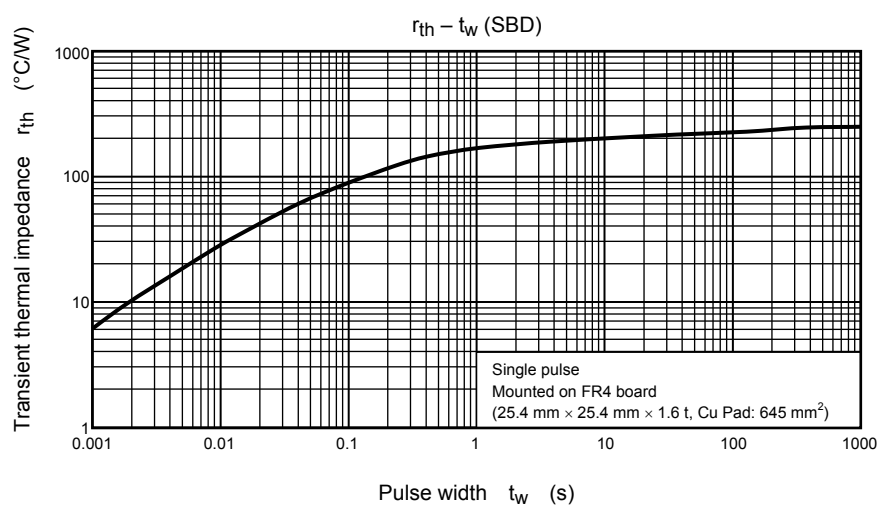
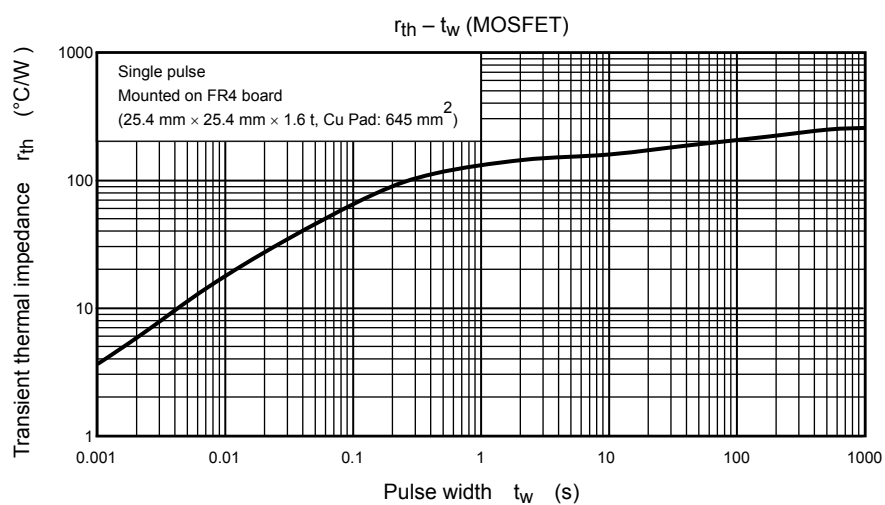




SBD Electrical Characteristics Graph



Transient thermal impedance Graph



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20070701-EN GENERAL

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